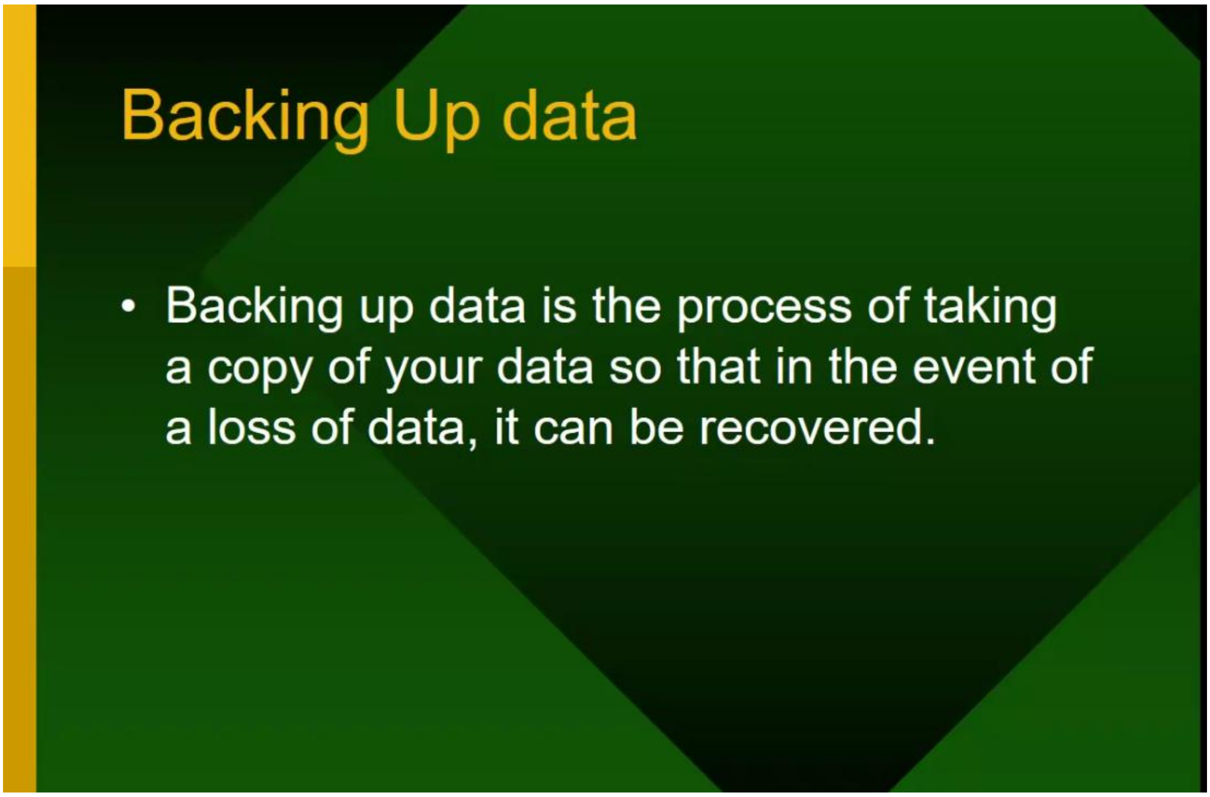
Intro to Backup and Backup Technology-56

1. Introduction to Backup

Backup refers to the process of creating a copy of data and storing it separately from the original source so that it can be restored in case of data loss, corruption, accidental deletion, or system failure. Backups are critical in IT infrastructure because data is the most valuable asset for any organization. Without backup, recovering from disasters like hardware failure, ransomware attacks, or human errors becomes impossible.

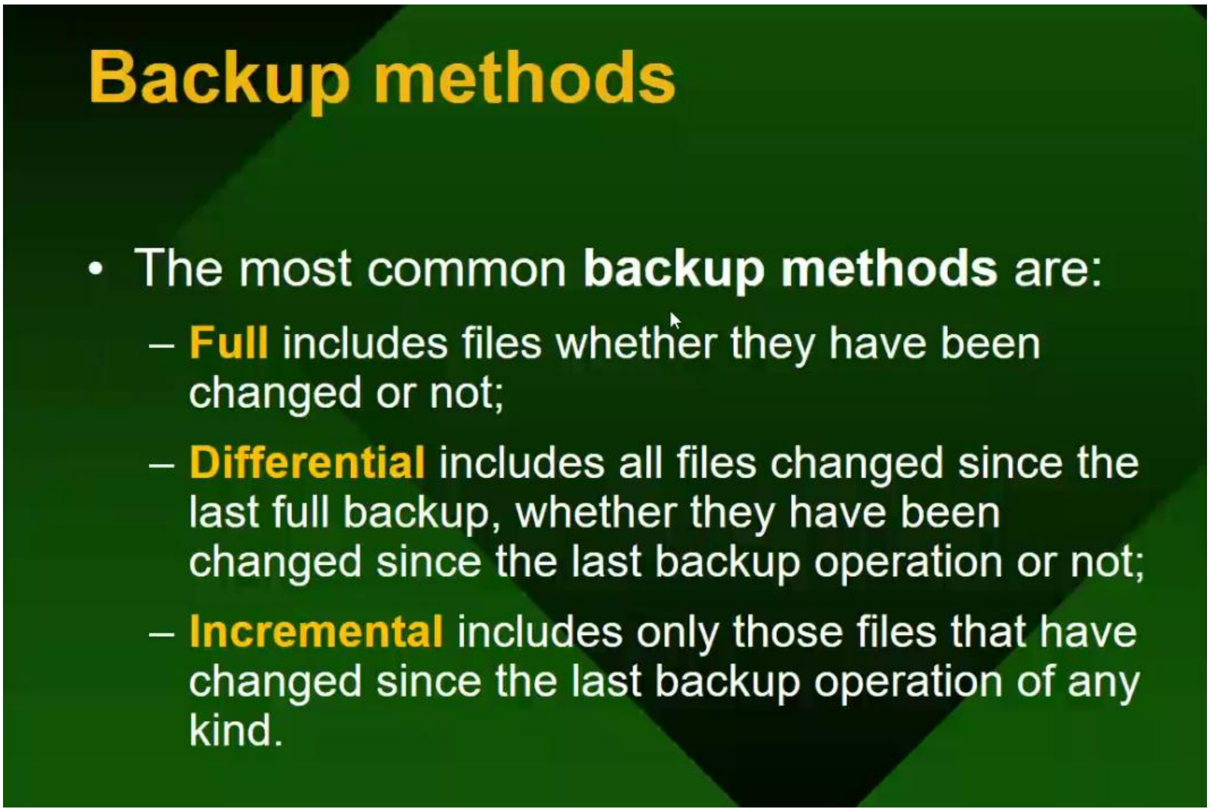


Key objectives of backup:

- Ensure data availability and integrity.
- Provide business continuity during downtime.
- Enable disaster recovery.
- Meet compliance and legal requirements.

2. Types of Backups

There are several methods of backup, each serving different needs:



1. **Full Backup**
    - Copies all selected data in one backup operation.
    - Pros: Simple, complete data set available.
    - Cons: Time-consuming, requires large storage.
2. **Incremental Backup**
    - Backs up only the data that has changed since the last backup (full or incremental).
    - Pros: Faster and uses less storage.
    - Cons: Restoration requires full backup plus all incremental backups.
3. **Differential Backup**
    - Backs up all data changed since the last full backup.
    - Pros: Faster restore than incremental.
    - Cons: Size grows with time until the next full backup.

3. Backup Technologies and Methods

Modern backup systems use multiple technologies:

- **On-Premises Backup**: Backup stored on local servers, NAS (Network Attached Storage), or SAN (Storage Area Network).
- **Cloud Backup**: Data is backed up to remote cloud storage (AWS Backup, Azure Backup, Google Cloud Backup).
- **Hybrid Backup**: Combination of local and cloud backup for flexibility.
- **Disk-to-Disk Backup**: Faster backup and restore compared to tapes.
- **Disk-to-Tape Backup**: Still used for long-term archival due to low cost.
- **Virtual Machine Backup**: Specialized tools for backing up entire VM environments (VMware, Hyper-V).

4. Backup Storage Media

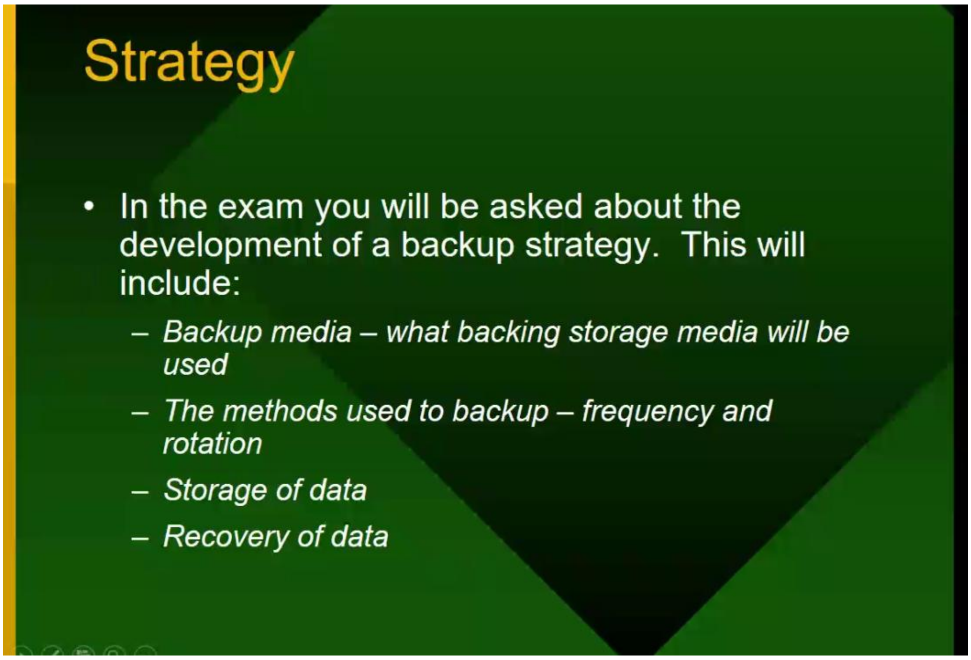
Backup media are the physical devices or storage systems where backup data is stored. The choice of media depends on factors like cost, speed, durability, and capacity.

- Digital Audio Tapes (DAT): Commonly used in enterprises because they are cheap, reliable, and can store large volumes of data.
- External Hard Drives: Easy to use, portable, and provide faster data access. Popular for personal and small business backups.
- Compact Discs (CDs): Earlier widely used for small data backups, though limited capacity makes them less practical today.
- Magnetic Tapes: Still used in data centers for long-term archival because of their low cost per gigabyte.
- Solid State Drives (SSDs): Faster and more durable than hard drives, but more expensive.
- Cloud Storage: Increasingly preferred due to scalability, accessibility, and reduced need for physical handling.

5. Backup Strategies

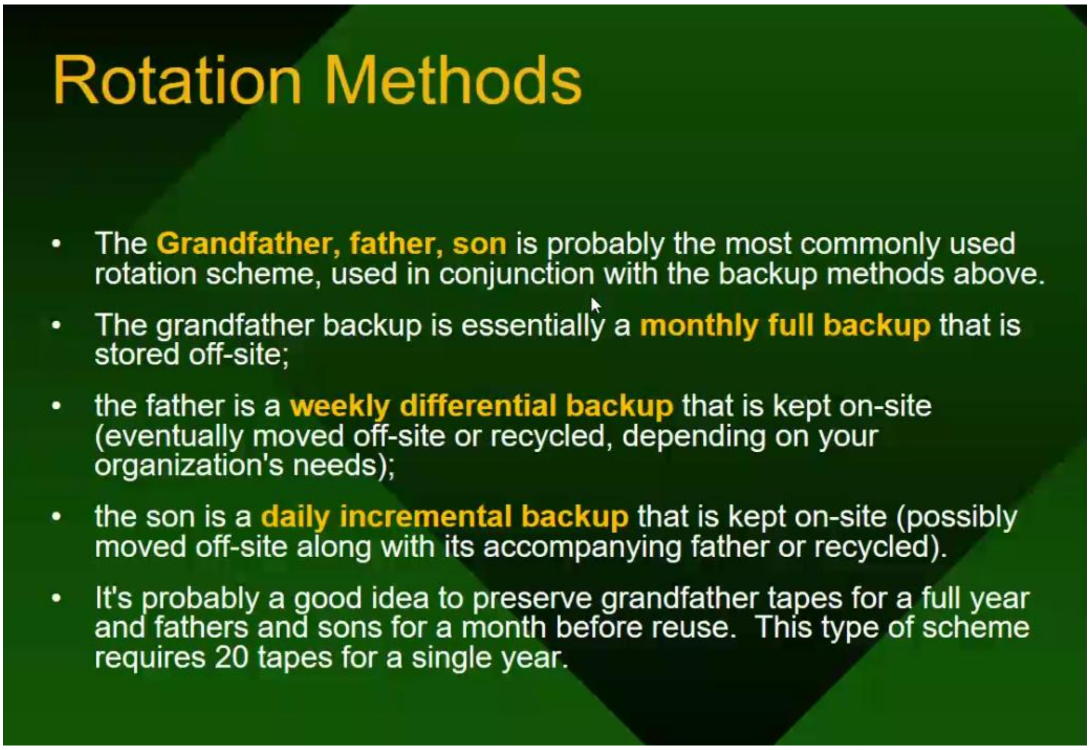
Organizations adopt strategies based on Recovery Point Objective (RPO) and Recovery Time Objective (RTO):

- **3-2-1 Backup Rule**: Keep 3 copies of data, on 2 different media, with 1 stored offsite.
- **Grandfather-Father-Son (GFS)**: Rotational backup system with daily, weekly, and monthly backups.
- **Tower of Hanoi**: Complex rotational scheme to balance space and data retention.



6. Challenges in Backup

- Increasing data volume leading to high storage cost.
- Backup window (time available for backup operations).
- Ensuring data consistency for live databases and applications.
- Protecting backup data from ransomware attacks.
- Compliance with regulations (GDPR, HIPAA).



7. Modern Trends in Backup Technology

- **Deduplication**: Eliminates duplicate data to save storage space.
- **Compression**: Reduces file sizes to optimize storage.
- **Encryption**: Ensures security of backup data.
- **AI-driven Backup**: Predicts failures and automates backup schedules.
- **Immutable Backups**: Backup data that cannot be altered, protecting against ransomware.

Backup is the foundation of IT resilience. With growing data and cyber threats, organizations are moving toward hybrid and cloud-based backup solutions. An effective backup strategy ensures business continuity, protects critical data, and supports compliance needs.